2_-_Exploratory-Data-Analysis.R

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Thu Mar 28 02:43:03 2019

```
################################################################################
# Cricket Health Technical Assessment
# Medicare Claims Analysis
# EDA file to stimulate hypothesis generation and understand pieces of the data
################################################################################
source(here::here("Cricket-Health-Technical-Assessment", "0 - Config.R"))
```

```
## ── Attaching packages ─────────────────────────────────────── tidyverse 1.2.1
──
```

```
## ✔ ggplot2 3.1.0     ✔ purrr   0.3.0
## ✔ tibble  2.0.1     ✔ dplyr   0.7.8
## ✔ tidyr   0.8.2     ✔ stringr 1.3.1
## ✔ readr   1.3.1     ✔ forcats 0.3.0
```

```
## ── Conflicts ────────────────────────────────────────── tidyverse_conflicts()
──
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()    masks stats::lag()
```

```
##
## Attaching package: 'assertthat'
```

```
## The following object is masked from 'package:tibble':
##
##     has_name
```

```
##
## Attaching package: 'lubridate'
```

```
## The following object is masked from 'package:base':
##
##     date
```

```
## here() starts at /home/kmishra9/Individual Projects
```

```
##
## Attaching package: 'here'
```

```
## The following object is masked from 'package:lubridate':
##
##     here
```

```
################################################################################
# Import cleaned data
################################################################################

beneficiary_combined_limited = read_rds(path = beneficiary_combined_path)
beneficiary_claims = read_rds(path = merged_data_path)

################################################################################
# Generate EDA report of merged dataset
################################################################################

configure_report(add_plot_prcomp = NULL, plot_prcomp_args = NULL)
```

```
# This actually took forever to run because of how big the data is (and I'm on a superco
mputer rn) so we'll report on a sample for now
create_report(data = beneficiary_claims %>% slice(1:nrow(beneficiary_claims) %>% sample
(size = 10000)),
              output_file = "beneficiary_claims_report.html",
              report_title = "Medicare Beneficiary Claims (2008-2010)")
```

```
##
##
## processing file: report.rmd
```

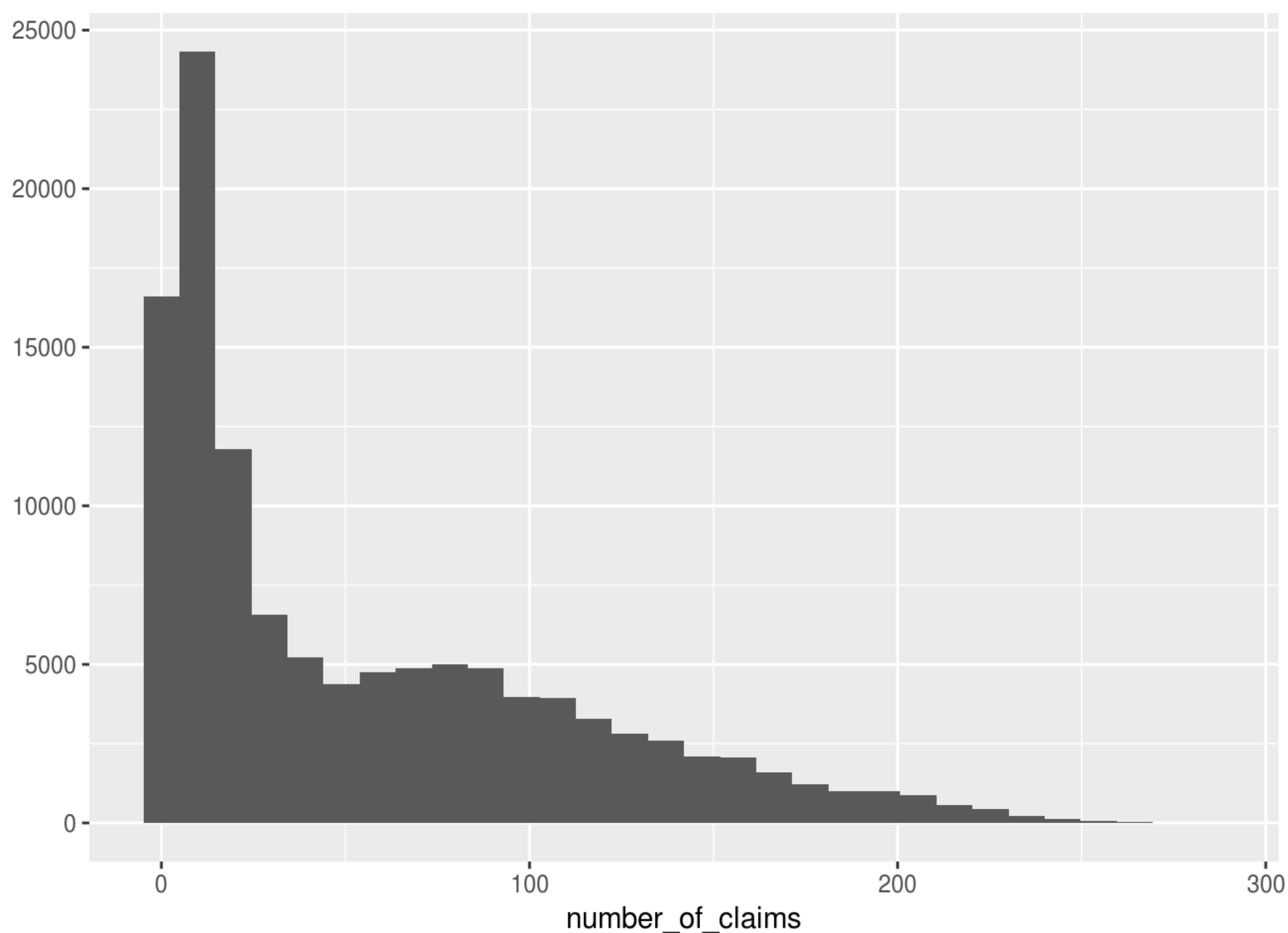
```
##
  |
  |                                                                 |   0%
  |
  |..                                                               |   2%
##    inline R code fragments
##
##
  |
  |...                                                              |   5%
## label: global_options (with options)
## List of 1
##  $ include: logi FALSE
##
##
  |
  |.....                                                            |   7%
##   ordinary text without R code
##
##
  |
  |......                                                           |  10%
## label: introduce
##
  |
  |........                                                         |  12%
##   ordinary text without R code
##
##
  |
  |.........                                                        |  14%
## label: plot_intro
```

```
################################################################################
# EDA on Specific Questions
################################################################################

# What proportion of beneficiaries are present over all three years?
beneficiary_combined_limited %>%
  group_by(beneficiary_id) %>%
  summarize(present_three_years = n() == 3) %>%
  pull(present_three_years) %>%
  sum / (beneficiary_combined_limited %>% pull(beneficiary_id) %>% unique %>% length)
```

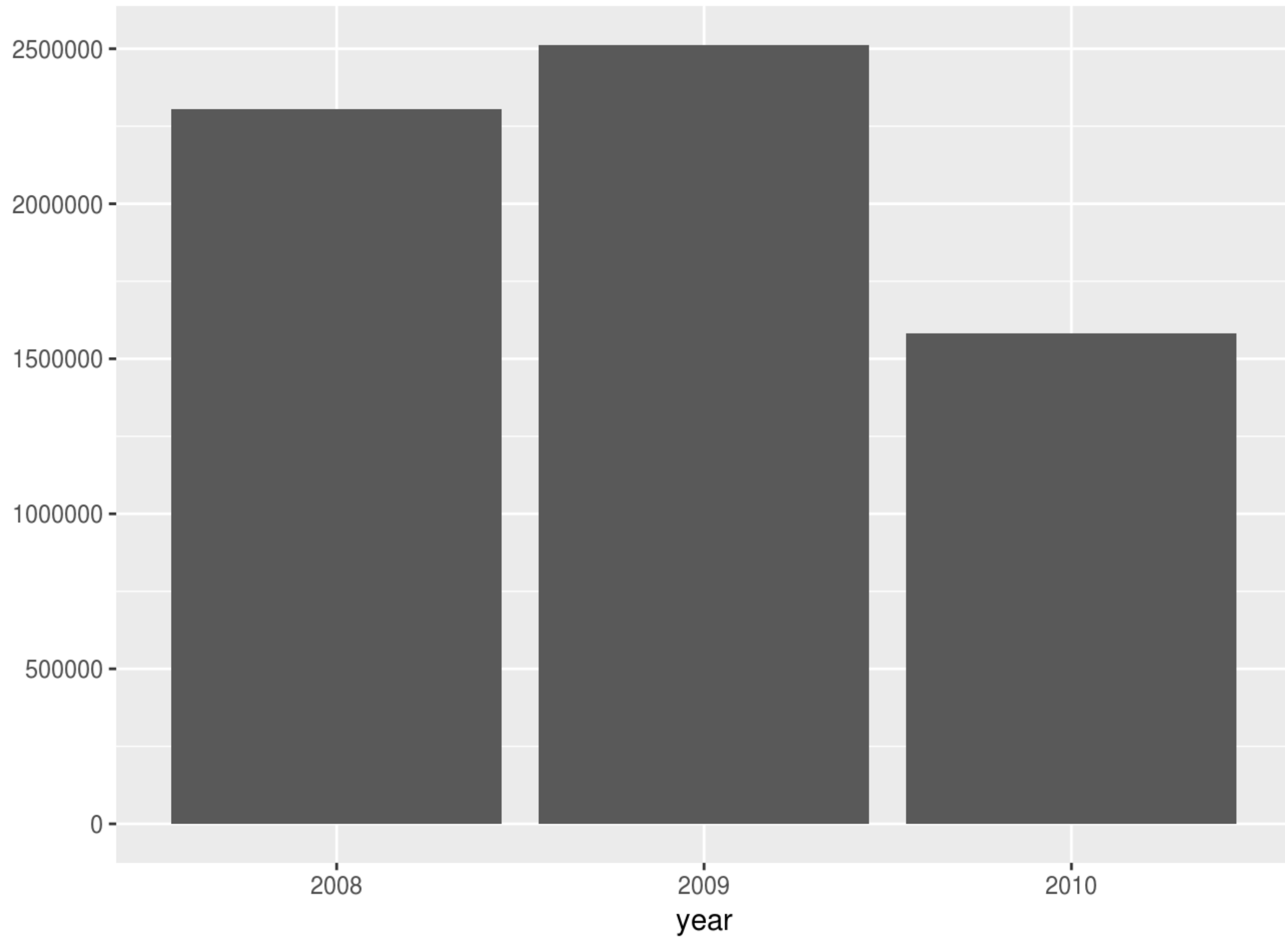
```
## [1] 0.9690766
```

```
# What does the distribution of # of claims per beneficiary look like?
num_claims = beneficiary_claims %>%
  group_by(beneficiary_id) %>%
  summarize("number_of_claims" = n())

qplot(x = number_of_claims , data = num_claims, geom = "histogram")
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



```
# How much heterogeneity in the number of claims from year to year is there?
qplot(
  x = year,
  data = beneficiary_claims %>% filter(!is.na(claim_payment_amount)) %>% mutate("year" =
as.factor(year)),
  geom = "bar"
)
```




```
# What proportion of beneficiaries are enrolled in part D? Over time, does this change?
 Is that at all related to the number of PDE claims?
beneficiary_combined_limited %>%
  filter(plan_d_coverage_length != "00") %>% nrow / (beneficiary_combined_limited %>% nr
ow)
```

```
## [1] 0.7597456
```

```
beneficiary_combined_limited %>%
  group_by(year) %>%
  summarize("proportion_on_plan_d" = sum(plan_d_coverage_length != "00")/ n())
```

```
## # A tibble: 3 x 2
##    year proportion_on_plan_d
##   <dbl>                <dbl>
## 1  2008                0.594
## 2  2009                0.808
## 3  2010                0.882
```

```
# More questions that I'd usually love to answer but I already have a feeling this is ov
erkill

# Are the Medicare reimbursement amounts correlated with beneficiary responsibility amou
nts and/or the number of claims a beneficiary makes?

# Are longer claims correlated with larger payment amounts?

# What was the longest claim length for each type of claim (just curious)

# Is end stage renal disease correlated with increased claims?

################################################################################
# Generate PDF Report of this script for later review of results
################################################################################

# NOTE: A report of this script has been generated and left in the Reports directory
```